



Mark Scheme (Results)

January 2026

Pearson BTEC National Level 3

Applied Science/Forensic & Criminal Investigation

Unit 1: Principles and Applications of Science I –
Biology (31617H/1B)

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Publication Code 31617H_1B_2601_MS

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Question Number	Answer	Additional Guidance	Mark
1(a)	B outer membrane (1)		(1)
1(b)	Pink/red (1)		(1)
1(c)	C nucleoid (1)		(1)
1(d)	Any two from: • (Antibiotics) {destroy/slow down the growth of} bacteria (1) • Disrupting {an organelle/ structure} (1) • Disrupting a biosynthetic pathway (1)	Accept a named organelle, such as the cell wall or cell membrane Accept a named pathway, such as protein synthesis, nucleic acid synthesis	(2)
Question Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)	• Difference (1) $86 - 15$ • Substitution (1) $\frac{71}{15} \times 100$ • Evaluation (1) $473.3 (\%)$ OR • Substitution (1) $\frac{86}{15} \times 100$ • Difference from 100 (1) $573.3 - 100$ • Evaluation (1) $473.3 (\%)$	Correct answer with no working gains full marks Accept 71 ECF Accept correct rounded answer to appropriate number of significant figures Accept 573.3 ECF Accept correct rounded answer to appropriate number of significant figures	(3)

[illegible]

Question Number	Answer	Additional Guidance	Mark
3(a)	(Simple) squamous epithelium (1)	Accept squamous	(1)
3(b)	Any two from: • (Persistent) cough (1) • <u>Increased</u> {mucus/phlegm} (1) • Breathlessness/difficulty breathing/lower blood oxygen levels (1) • <u>Increased</u> respiratory infections (1) • Wheezing (1) • Chest pain (1) • Tiredness/fatigue (1) • Swollen ankles (1) • Weight loss (1) • {Inflamed/narrowed} airway (1)		(2)

3(c)	- Decreased surface area for gas exchange (1) (Therefore) less oxygen to the {blood/muscles/cells/mitochondria} (1) (Therefore) <u>less</u> aerobic respiration is possible (1) <u>Less</u> ATP produced (making exercise difficult) (1)	Accept decreased surface area for (the diffusion of) oxygen Accept <u>less</u> energy released	(4)
Question Total = 7 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	- Allows the cells to be more flexible (1) - Which allows for red blood cells to pass through {narrow vessels/ capillaries} (1)		(2)
4(b)	Any four from: - As partial pressure of oxygen increases the percentage of haemoglobin saturation increases (1) - Haemoglobin has four (oxygen) binding sites (1) - Less oxygen binds at <u>low</u> PO_2/ it is hardest for the first oxygen molecule to bind (1) - The first oxygen molecule that binds causes a conformational change that allows the next two oxygen molecules to bind more easily (1) (Therefore) a <u>small</u> increase in PO_2 causes a large increase in haemoglobin saturation (1) - It is hard to bind the final oxygen molecule as most binding sites are occupied (1)	ORA Reject a linear relationship/ directly proportional Accept description of the increased affinity for oxygen binding	(4)
Question Total = 6 marks			

Question Number	Indicative Content	Mark
5	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> • Prokaryotic cells are unicellular • Prokaryotic cells can have flagellum • Euglena is a eukaryote and not a prokaryote • It has a nucleus • Eukaryote DNA is housed in a clearly defined nucleus with a nuclear membrane • It has membrane bound organelles, e.g. mitochondria, Golgi apparatus • Euglena could be characterised as either a plant or animal cell • Euglena could be a plant cell because • It contains chloroplasts • Plant cells contain chloroplasts • Animal cells do not contain chloroplasts • Plants photosynthesise • Euglena could be an animal cell because • It does not have a cell wall • Animal cells do not have a cell wall • Plant cells do have a cell wall • The large vacuole in plant cells is not contractile • Plant cell are not motile.	(6)

Level	Mark	Descriptor – Discuss All traits carry equal weighting
0	0	No rewardable material.
1	1-2	- Adequate interpretation, analysis and/or evaluation of the scientific information with generalised comments being made. - Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. - The discussion shows some structure and coherence.
2	3-4	- Good analysis, interpretation and/or evaluation of the scientific information. - Lines of argument mostly supported through the application of relevant evidence. - The discussion shows a structure which is mostly clear, coherent and logical.
3	5-6	- Comprehensive analysis, interpretation and/or evaluation of all pieces of scientific information. - Line(s) of argument consistently supported throughout by sustained application of relevant evidence. - The discussion shows a well-developed structure which is clear, coherent and logical.
Question Total = 6 marks		